RETAILIQ · AI BUILDER CUP 2026

Sensing. Execution. Empowerment.

A field execution copilot that turns shelf evidence into an accountable retail action.

Primary user: the CPG field representative.
Managers review. Buyers approve. Shoppers discover.

A. TEAM NAME

RetailIQ

B. TEAM LEADER AND MEMBERS

Abhishek Agarwal · Team leader

Vivek Sharma · Team member

C. PROBLEM STATEMENT

Retail & Commerce: Intelligent Customer and Business Experiences

Build an AI-powered solution that transforms how businesses understand, engage and serve customers while improving operational efficiency across physical and digital commerce.

THE FIELD EXECUTION COPILOT

See the gap. Know the action. Empower the rep.

One retail visit connects shelf evidence, a stock-aware decision and a human who can act.

Sensing → **Execution** → **Empowerment**

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EVIDENCE BECOMES A USEFUL NEXT STEP



Real shelf reference · fictional Protiva commercial scenario

Observe → **Verify** → **Save** → **Review**

From an audit result to an accountable visit record.

DIFFERENCE · PROBLEM SOLVING · USP

Close the gap between detection and action.

HOW THE IDEA DIFFERS

A complete visit, not an isolated score.

The same workflow carries shelf evidence, inventory context, a prepared conversation and a saved manager handoff.

Design comparison with isolated vision, analytics or chat workflows; no exhaustive competitor benchmark is claimed.

HOW IT SOLVES THE PROBLEM

Verify the stock state before buying more.

Zero facings plus 64 recorded units triggers a locate-and-reconcile task. Demand, inbound stock, cases and shelf life constrain a justified proposed order.

The 64-unit scenario is synthetic. System records require a physical check.

USP OF THE PROPOSED SOLUTION

Evidence + retail logic + human authority.

Own-brand and competitor questions stay separate. Unseen evidence stays unknown. Source-linked explanations and a saved record let the rep and manager inspect the reasoning.

Research rationale: a 55-retailer study reports out-of-stocks around 8% and links performance to aligned sensing, execution and empowerment. We apply that finding as a design principle, not as proof of RetailIQ impact.

Source: Corsten, Gruen & LaBrecque, IJRDM, 28 May 2026. Public abstract; fieldwork dates are not disclosed.

IMPLEMENTED FEATURES BY USER

Field execution, review and product discovery.

FIELD REPRESENTATIVE

Sense and execute

- Still/video shelf evidence and own-brand audit
- Separate competitor execution benchmark
- Guided counts and stock discrepancy alerts
- Forecast-aware suggested orders
- Source-linked pitch, buyer rehearsal and debrief

FIELD MANAGER AND BUYER

Review and control

- Saved audits and proposed order records
- Revenue, forecast intervals and inventory alerts
- Customer segment context and team reference views
- Optional private photo retrieval in cloud mode
- Buyer retains commercial approval

SHOPPER

Discover and compare

- Conversational catalogue discovery
- Prices, nutrition and stock context
- Preferences and basket-aware assistance
- Available product comparisons
- Clearly labelled synthetic commercial fixture

ONE VISIT · TRACE THE DECISION

Observe → verify → act → review.

01 · SENSE

Capture shelf

Photo or selected video frames; preserve the visit context.

→

02 · QUALIFY

Assess visibility

PASS, FAIL or NOT VISIBLE; human confirms counts.

→

03 · VERIFY

Check stock

Compare shelf evidence with system and inbound quantities.

→

04 · EXECUTE

Choose action

Locate/reconcile stock, or size justified replenishment.

↓

05 · EMPOWER

Prepare the dialogue

Inspect source-linked facts; rehearse the buyer conversation.

←

06 · HUMAN CONTROL

Review the draft

Rep and buyer validate counts, terms and commercial commitment.

←

07 · RECORD

Save the visit

Audit, proposed action and debrief; local or configured cloud.

←

08 · REVIEW

Manager handoff

Inspect the record and follow up on the unresolved action.

Decision branch: empty shelf + positive recorded stock → locate and reconcile first. Verified replenishment need → demand minus usable stock and inbound, constrained by cases and shelf life.

SCREEN STRUCTURE · EXPLANATORY MOCK DIAGRAMS

Three surfaces, one commercial context.

Field Execution

- Visit / store / buyer context
- Shelf evidence + qualified audit verdicts
- Confirmed counts
- Stock logic
- **Review next action → save visit**

Team and Analytics

- Revenue
- Inventory risk
- Forecast intervals + segment context
- Saved audit / proposed order events
- **Inspect evidence → follow up**

Shopper Insights

- Need, preferences and basket context
- Conversation and available options
- Price / nutrition
- Catalogue stock
- **Compare suitable products**

These diagrams explain the implemented layout. Actual prototype screenshots follow in the required snapshot section.

BUILT FOR THE CHALLENGE

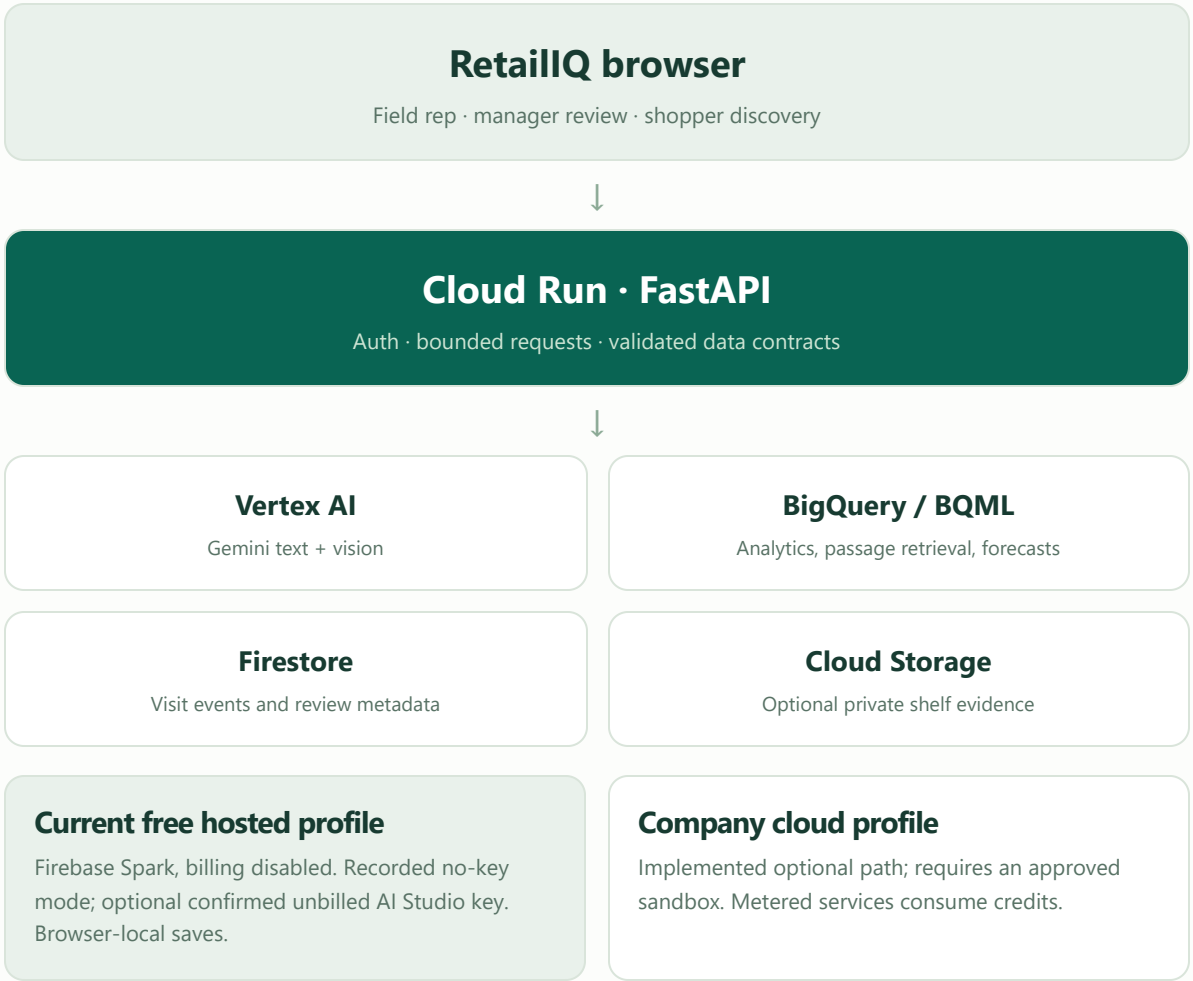
Current free host. Implemented backend reference.

Current path: Firebase Spark serves the browser; optional unbilled AI Studio access is direct. The diagram shows the separate implemented backend reference, historically verified.

Inventory intelligence	Stock, inbound and shelf-life decisions
Demand planning + customer insight	BQML, analytics and segment context
Conversational product discovery	Catalogue-grounded assistance

Private access by default. Explicit shared-demo auth. Optional retained evidence stays private.

IMPLEMENTED BACKEND REFERENCE · NOT THE CURRENT STATIC HOST



TECHNOLOGY CHOICES · PURPOSE AND ACTUAL STATUS

Google services with explicit boundaries.

Technology	Role in the solution	Status
Firebase Hosting · Spark	Public static app, presentation and video player	Current host; project billing disabled
HTML / CSS / JavaScript	Rep, manager and shopper surfaces; browser-local events	Current application
Gemini / Google AI Studio	Optional direct text and vision assistance	Unbilled-key confirmation; free limit stops calls
Cloud Run / FastAPI	Authenticated API and frontend serving	Implemented backend reference
Gemini on Vertex AI	Backend text and shelf vision; service-account access	Historical 3.6 Flash reference evaluated
BigQuery / BigQuery ML	Commercial analytics, passage retrieval, ARIMA forecasting	Implemented; synthetic history tested
Firestore	Saved visit events and review metadata	Implemented cloud persistence
Cloud Storage / Secret Manager	Optional private evidence and server credential storage	Implemented backend safeguards
Playwright / pytest / GitHub Actions	Browser, API, deployment and evaluator regressions	Local checks and CI workflow supplied

No usage of Vertex AI Search, Looker, AlloyDB or Cloud SQL is claimed. Current free-tier model quality has not been live-evaluated.

COST AND VALUE · EXPLICIT PLANNING ASSUMPTIONS

Put a value on productive time.

154 hours/month capacity

10 reps × 6 visits/day × 22 days	**1,320 visits**
Scoped task: 12-minute baseline → 5-minute target	**7 min/visit**
154 hours × assumed ₹300/hour	**₹46,200**
Target ₹5/visit operating budget	**₹6,600**
Illustrative net capacity value	**₹39,600/month**

Scope: audit preparation and debrief administration. Not the entire visit. All inputs require pilot measurement.

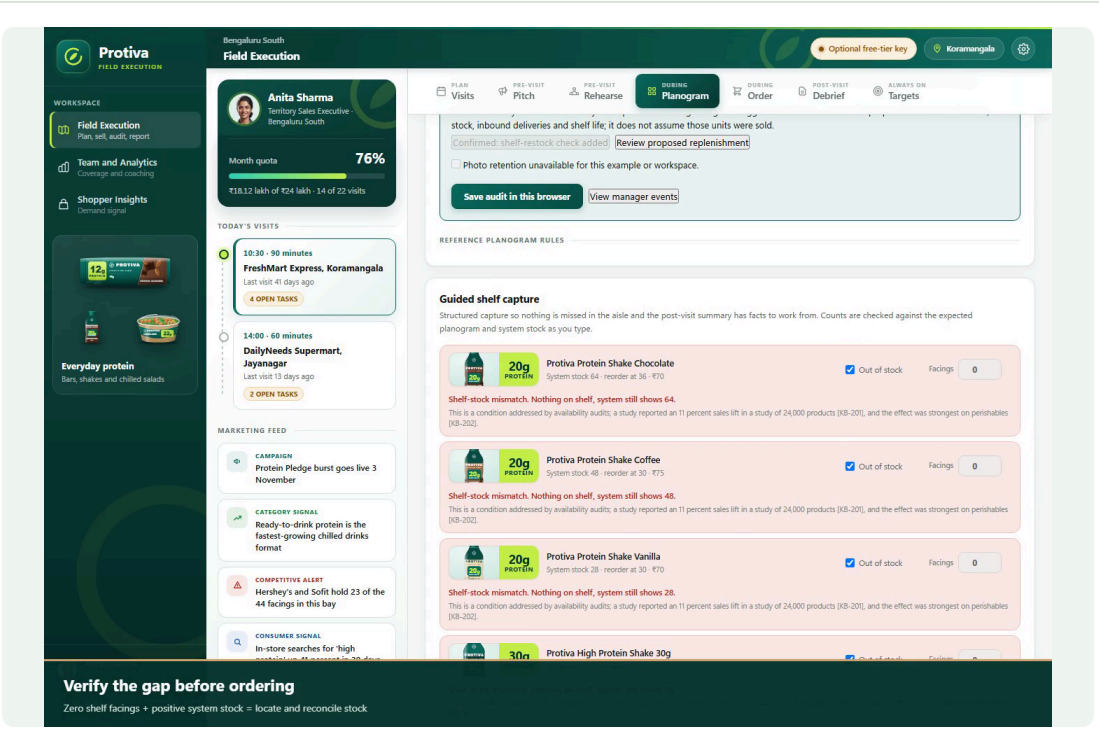
Cost profile	What it means
Local / Firebase Spark	Recorded mode makes no AI requests. Unbilled static hosting is quota bounded; billing stays disabled. Saves remain in the browser.
Optional free AI	Use an unbilled AI Studio project with available quota. Quota exhaustion stops requests; no paid fallback.
Company credits	Cloud Run, Vertex AI, BigQuery and storage are metered. Usage consumes credits. Actual credit balance and total cost remain unverified.

Capacity is not guaranteed cash saving. Value depends on how released time is used. Implementation, integration, training and devices are excluded.

Implementation cost is not yet estimated: integration, training and device scope require discovery. The operating budget is a target, not an invoice. No achieved uplift is included.

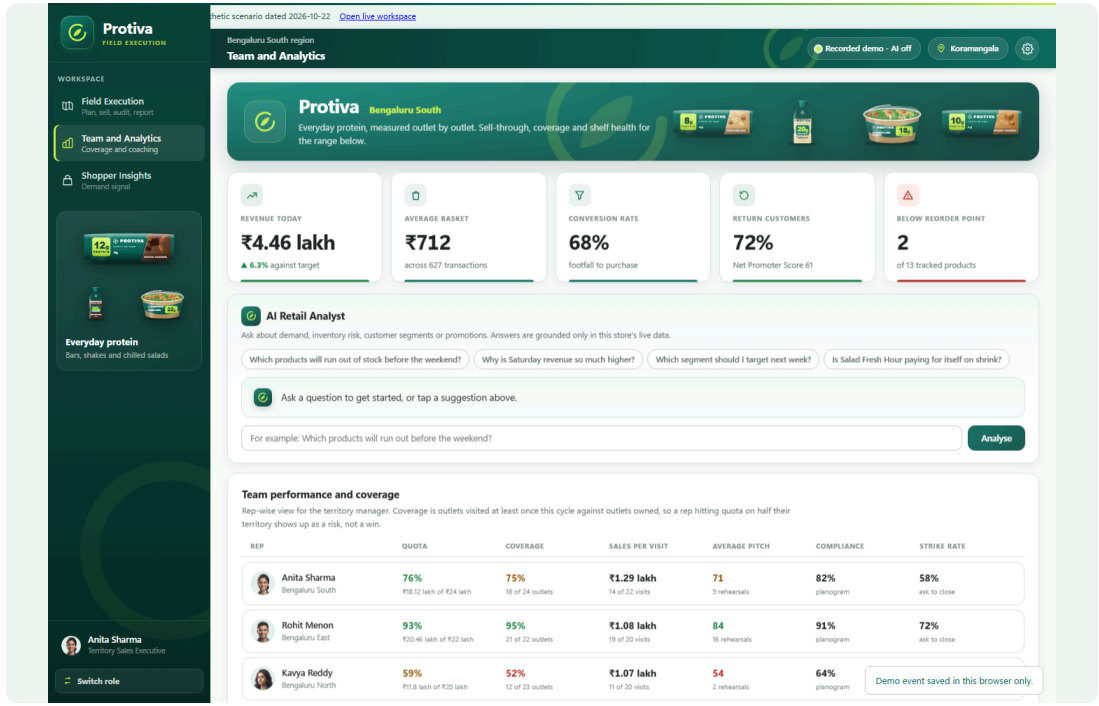
ACTUAL APPLICATION CAPTURES · NO FABRICATED PRODUCT MOCKUPS

A working rep workflow and manager view.



Field rep: confirm the shelf-stock mismatch

Real prototype screen. The synthetic example separates zero facings from 64 recorded stock units and saves an accountable next action.



Manager: inspect commercial context and records

Real prototype screen. Commercial KPIs, team profiles and history are synthetic. Current free hosting saves events in the browser.

SOFTWARE, MODEL REFERENCE AND BUSINESS IMPACT ARE SEPARATE

Verification with comparable baselines and limits.

Area	Measured check / result	What it establishes and does not
Current public static host	18 HTTP/browser checks passed	Asset hashes, private-path rejection, mobile fit and local visit workflow; zero AI/backend requests
Public video player	6 checks passed; native Chromium playback	Correct MIME, published hashes, 222.44-second playable MP4; zero AI/backend requests
Browser + policy regressions	157 browser + 7 policy checks passed	Exercised UI contracts with mocked backend responses
Python regressions	60 passed: 51 API + 4 deployment + 5 free-hosting guards	Cloud doubles; not a new backend deployment
Offline evaluator	38 deterministic assertions passed	Known facts and rejection of corrupted responses; not model-quality scoring
Historical live model reference	10/10 synthetic cases; Gemini 3.6 Flash on Vertex AI	Small historical paid-reference set; does not validate the current free-tier model
Historical cloud reference	22 deployed cloud checks passed	Warehouse, protected events and private photos; that endpoint is retired
Synthetic forecast holdout	7 days; WAPE 1.249% vs seasonal-naive 2.824%	Diagnostic improvement on generated data; not real-store forecast accuracy

Reports: submission/evidence/. Business uplift, visual counting precision and production cost remain pilot hypotheses.

ROADMAP · PROPOSED, NOT ALREADY DELIVERED

Prove the decision. Then scale the integration.

01 Operational data integration

Replace synthetic history with permissioned POS, stock and inbound feeds; preserve provenance.

02 Independent visual evaluation

Label shelf observations and quantify precision, coverage and honest abstention across stores.

03 Enterprise handoff

Optional approved company backend, identity integration and ERP draft-order interfaces.

PILOT · FALSIFIABLE SUCCESS CRITERIA

Two-week baseline, then an assisted period.

Test the five-minute target for scoped audit preparation and debrief administration. Measure corrections and productive use of released time.

Correctness	Independent shelf labels; precision and abstention
Execution	Verified stock discrepancy closure; human review
Business viability	Observed cost, latency, waste risk and accepted actions

No achieved revenue uplift or stockout reduction is claimed. The optional current free-tier model still needs live quality evaluation. Company credits do not make metered services inherently free.

GITHUB · DEMO VIDEO · FINAL PRODUCT

Review the source. Watch the visit. Try the prototype.

All three submission destinations are listed explicitly. The current product is a free static recorded experience; commercial context is synthetic.

TEAM RETAILIQ

Abhishek Agarwal · Team leader

Vivek Sharma · Team member

Sensing + execution + empowerment, in one accountable visit.

GitHub Public Repository

github.com/abhishek-0706/AI-BUILDER-CUP-2026

Exact user-selected repository; public access verified without authentication.

Demo Video Link

retailiq-free-20261018-7ce297.web.app/demo.html

3:42.44 · AI-assisted creator-voice narration · recorded scenario. Meets the form's 3–4-minute range; the template labels this field “3 Minutes”.

Final Product Link

retailiq-free-20261018-7ce297.web.app/?demo=recorded

Firebase Spark, billing disabled. No key or model requests required. Events save in the browser.